

Configuration Automation using Ansible

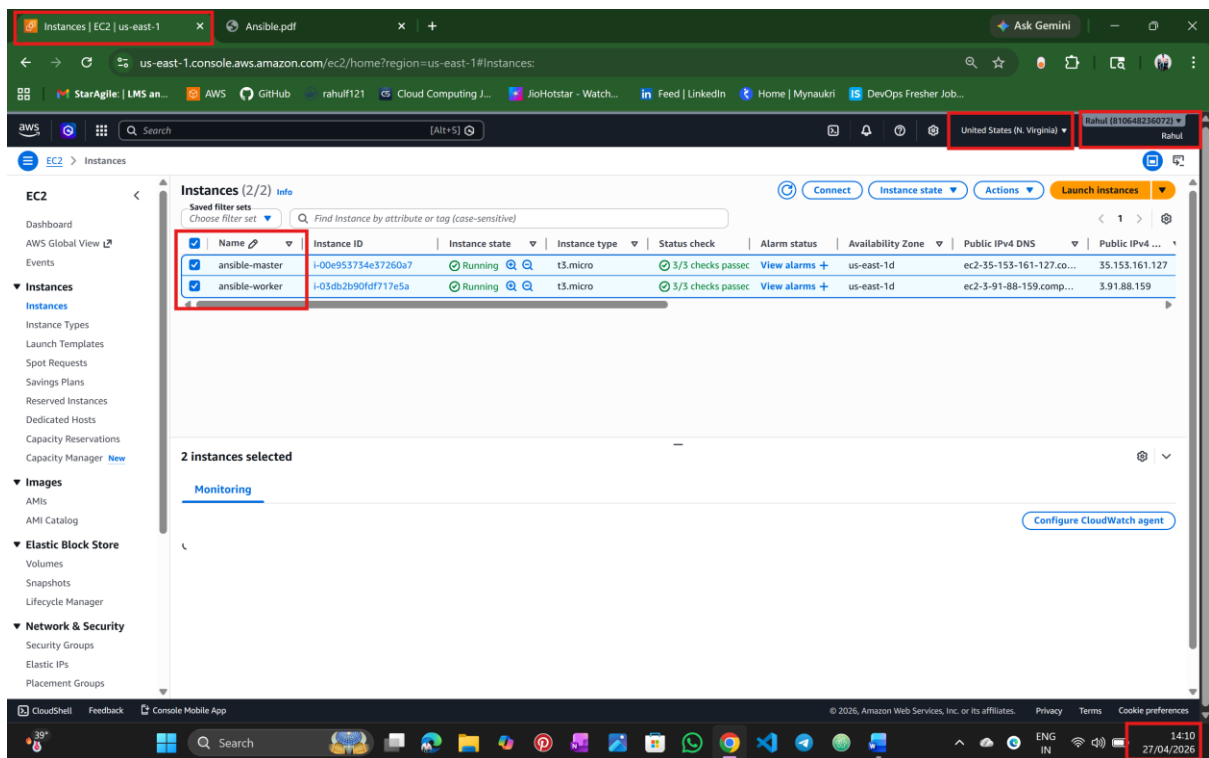
Submitted to - Vikul Sir

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L2 - Create and Execute Ansible Playbook to Install Docker and Run the Docker Application Image created in Docker Module

Step:1 login into AWS console and create ec2 instance



Step:2 This is ansible master node

The screenshot shows the AWS EC2 Instance Connect console for an instance named 'i-00e953734e37260a7 (ansible-master)'. The terminal window displays the output of the 'ansible --version' command. The output shows the ansible version as 2.15.3, the python version as 3.9.25, and the jinja version as 3.1.4. The terminal also shows the ansible configuration file location and the ansible python module location. The terminal output is as follows:

```
last login: Sun Apr 26 08:58:45 2026 from 18.206.107.20
[ec2-user@ip-172-31-47-209 ~]$ sudo su
[root@ip-172-31-47-209 ec2-user]# ansible --version
ansible [core 2.15.3]
  config file = None
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.9/site-packages/ansible
  ansible collection location = /root/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.9.25 (main, Mar 13 2026, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (/usr/bin/python3.9)
  jinja version = 3.1.4
  libyaml = True
[root@ip-172-31-47-209 ec2-user]#
```

The terminal window also shows the public IP address of the instance: 35.153.161.127. The console shows the instance is in the 'United States (N. Virginia)' region and the user is 'Rahul'.

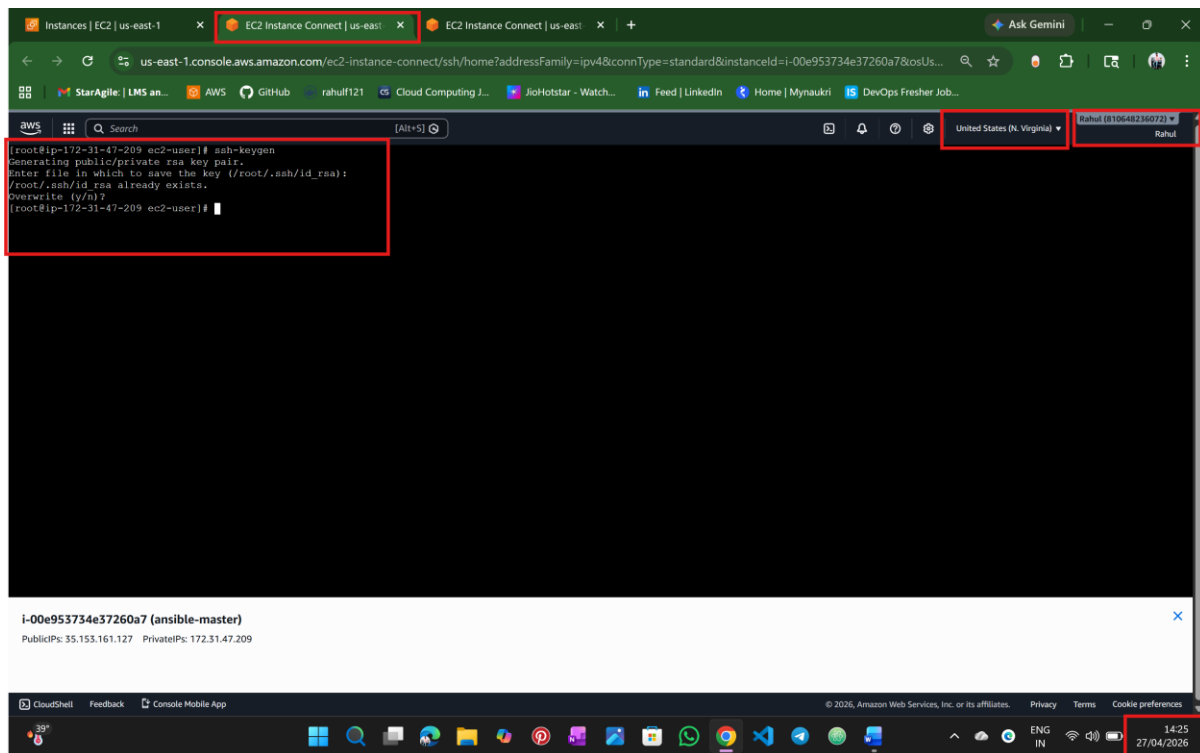
Step:3 This is ansible worker node

The screenshot shows the AWS EC2 Instance Connect console for an instance named 'i-03db2b90df717e5a (ansible-worker)'. The terminal window displays the output of the 'ansible --version' command. The output shows the ansible version as 2.15.3, the python version as 3.9.25, and the jinja version as 3.1.4. The terminal also shows the ansible configuration file location and the ansible python module location. The terminal output is as follows:

```
last login: Sun Apr 26 10:09:18 2026 from 172.31.47.209
[ec2-user@ip-172-31-38-210 ~]$ sudo su
[root@ip-172-31-38-210 ec2-user]# ansible --version
ansible [core 2.15.3]
  config file = None
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.9/site-packages/ansible
  ansible collection location = /root/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.9.25 (main, Mar 13 2026, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (/usr/bin/python3.9)
  jinja version = 3.1.4
  libyaml = True
[root@ip-172-31-38-210 ec2-user]#
```

The terminal window also shows the public IP address of the instance: 3.91.88.159. The console shows the instance is in the 'United States (N. Virginia)' region and the user is 'Rahul'.

Step:4 connect ansible master node and worker node with ssh-keygen, here I already connected to it.

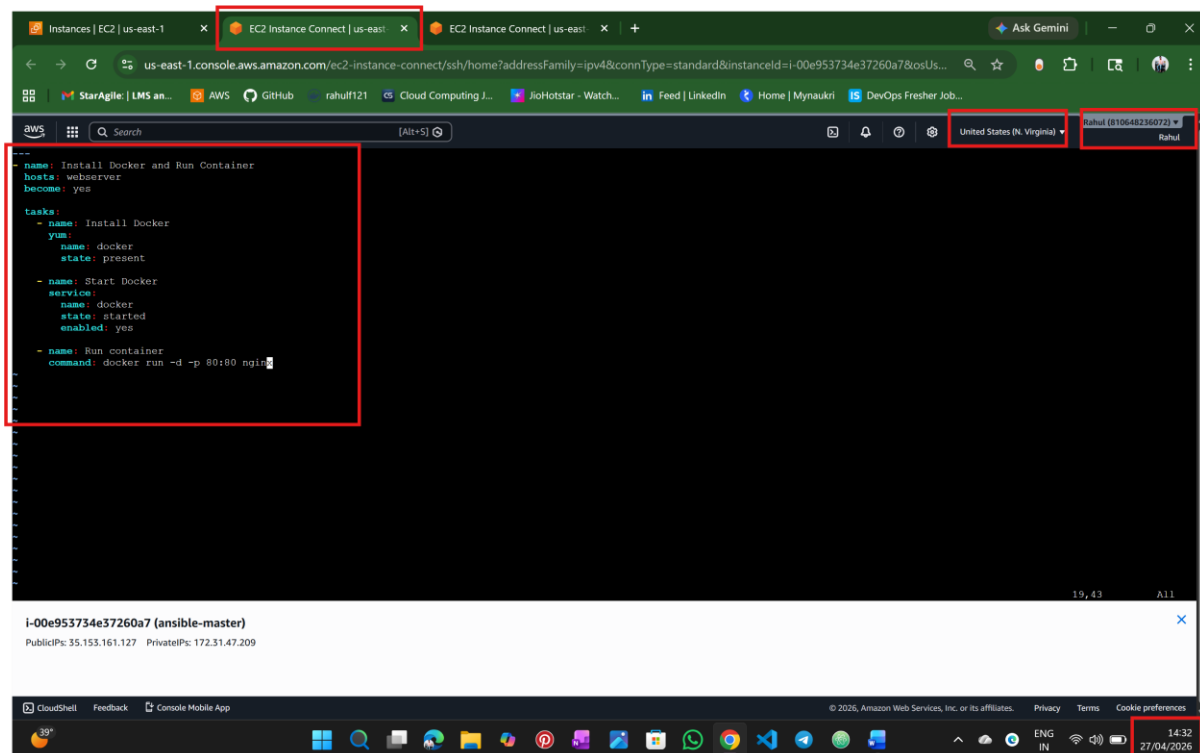


The screenshot shows the AWS CloudShell interface. The terminal window displays the following output for the `ssh-keygen` command:

```
[root@ip-172-31-47-209 ec2-user]# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
/root/.ssh/id_rsa already exists.
Overwrite (y/n)?
[root@ip-172-31-47-209 ec2-user]#
```

The interface also shows the instance details for `i-00e953734e37260a7 (ansible-master)` with Public IP: 35.153.161.127 and Private IP: 172.31.47.209. The top navigation bar indicates the region is `United States (N. Virginia)` and the user is `Rahul`. The bottom status bar shows the time as 14:25 on 27/04/2026.

Step:5 This is yaml file



The screenshot shows the AWS CloudShell interface with an Ansible playbook file. The terminal window displays the following content:

```
---
- name: Install Docker and Run Container
  hosts: webservers
  become: yes
  tasks:
    - name: Install Docker
      yum:
        name: docker
        state: present
    - name: Start Docker
      service:
        name: docker
        state: started
        enabled: yes
    - name: Run container
      command: docker run -d -p 80:80 nginx
```

The interface also shows the instance details for `i-00e953734e37260a7 (ansible-master)` with Public IP: 35.153.161.127 and Private IP: 172.31.47.209. The top navigation bar indicates the region is `United States (N. Virginia)` and the user is `Rahul`. The bottom status bar shows the time as 14:32 on 27/04/2026.

Step:6 verify and check the connect and run the yaml file

The screenshot shows the AWS Management Console interface for an EC2 Instance Connect session. The browser tabs at the top include 'Instances | EC2 | us-east-1', 'EC2 Instance Connect | us-east-1', and 'EC2 Instance Connect | us-east-1'. The address bar shows the URL: `us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-00e953734e37260a7&osUs...`. The AWS console header shows the region 'United States (N. Virginia)' and the user 'Rahul (810648236072)'. The terminal window displays the following commands and output:

```
[root@ip-172-31-47-209 /]# vi hosts
[root@ip-172-31-47-209 /]# ansible -i hosts webserver -m ping
[WARNING]: Platform linux on host 172.31.38.210 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.38.210 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[root@ip-172-31-47-209 /]# vi docker-playbook.yml
[root@ip-172-31-47-209 /]# ansible-playbook -i hosts docker-playbook.yml
PLAY [Install Docker and Run Container] *****
TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 172.31.38.210 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
172.31.38.210 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
TASK [Install Docker] *****
changed: [172.31.38.210]
TASK [Start Docker] *****
changed: [172.31.38.210]
TASK [Run container] *****
changed: [172.31.38.210]
PLAY RECAP *****
172.31.38.210 : ok=4 changed=3 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
[root@ip-172-31-47-209 /]#
```

Below the terminal window, the instance details for 'i-00e953734e37260a7 (ansible-master)' are shown, including Public IP: 35.153.161.127 and Private IP: 172.31.47.209. The bottom status bar shows the time as 14:35 on 27/04/2026.

Step:7 verify the container

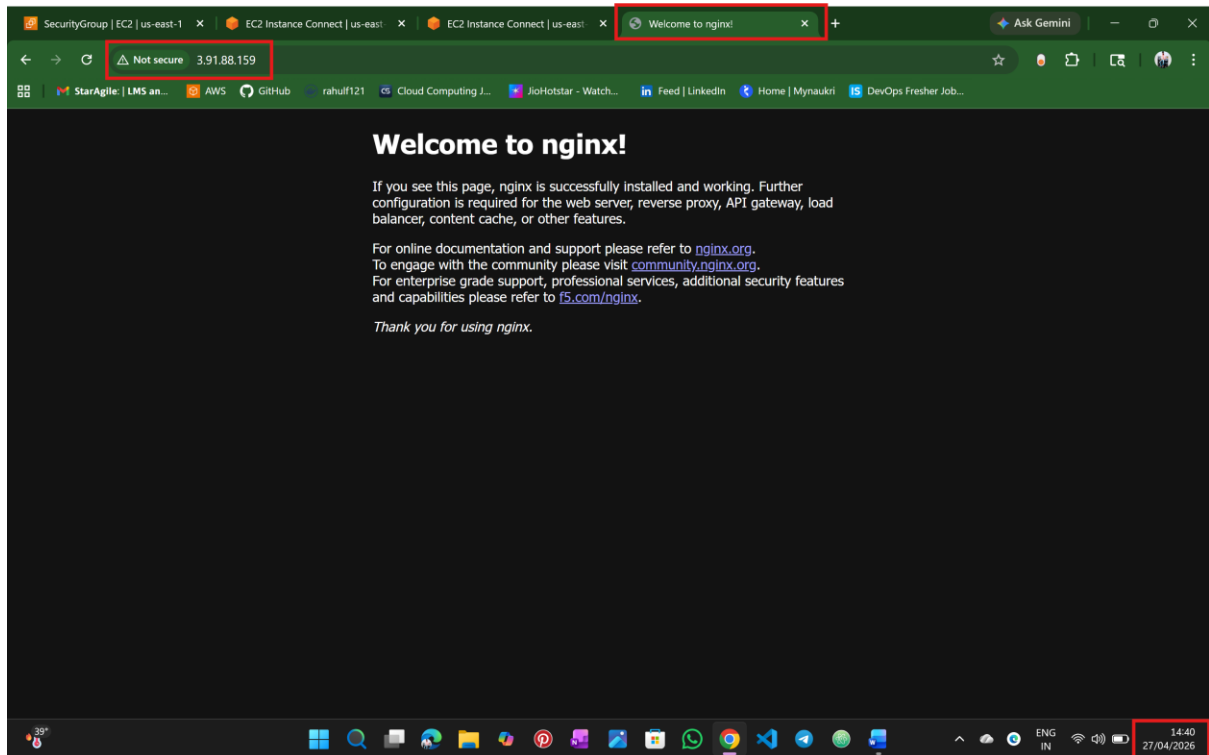
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The screenshot shows the AWS Management Console interface for an EC2 Instance Connect session. The browser tabs at the top include 'SecurityGroup | EC2 | us-east-1', 'EC2 Instance Connect | us-east-1', and 'EC2 Instance Connect | us-east-1'. The address bar shows the URL: `us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-1&connType=standard&instanceId=i-03db2b90fd717e5a&osUser=...`. The AWS console header shows the region 'United States (N. Virginia)' and the user 'Rahul (810648236072)'. The terminal window displays the following commands and output:

```
[ec2-user@ip-172-31-38-210 ~]$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
654d249cc84e   nginx    "/docker-entrypoint..." 3 minutes ago Up 3 minutes 0.0.0.0:80->80/tcp, :::80->80/tcp   condensing_varahamihira
[ec2-user@ip-172-31-38-210 ~]$ sudo usermod -s /bin/bash ec2-user
[ec2-user@ip-172-31-38-210 ~]$
```

Below the terminal window, the instance details for 'i-03db2b90fd717e5a (ansible-worker)' are shown, including Public IP: 3.91.88.159 and Private IP: 172.31.38.210. The bottom status bar shows the time as 14:39 on 27/04/2026.

Step:8 access the website using worker node ip address



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Nalathati Rahul